Atharva Mahajan

Generative Models & Reinforcement Learning for Physical Systems · M.S. University of Michigan

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Profile

Machine-learning researcher building **generative models and closed-loop control policies for high-dimensional physical systems**. First author of *Policy-DRIFT* - NeurIPS 2026 (conditional flow matching, reward-guided sampling, deep RL), which reaches **48.95% drag reduction at 37× lower actuation energy**; prior work spans neural surrogates, DNS/LES turbulence studies, and **450× CUDA acceleration** of a CFD solver.

Education

University of Michigan - ***M.S. Mechanical Engineering, GPA 3.98/4; focus: generative modeling, RL*** Aug 2025 – May 2027

BITS Pilani, Hyderabad - ***B.E. Mechanical Engineering (CGPA 9.16/10) · M.Sc. Physics*** Nov 2020 – May 2025

Technical Skills

Machine Learning and AI: Generative modeling (conditional flow matching, diffusion, guided sampling), deep reinforcement learning, reward modeling, self-supervised representation learning (JEPA), foundation models, CNNs/RNNs/autoencoders, neural surrogates & reduced-order modeling, physics-informed ML, closed-loop control
Languages & Frameworks: Python, C/C++, CUDA, Fortran, MATLAB, Bash · PyTorch, TensorFlow, NVIDIA Nsight, GPU/HPC clusters, Git, LaTeX
Simulation & Hardware: DNS/LES, spectral & meshfree solvers, boundary element methods, Dedalus, ANSYS Fluent, COMSOL, SolidWorks, Fusion 360, Arduino / embedded control

Research Experience: Machine Learning & Simulation

Policy-DRIFT: Dynamic Reward-Informed Flow Trajectory Steering Aug 2025 – Present
First author · *Graduate Researcher, Prof. Ricardo Vinuesa, University of Michigan · NeurIPS 2026* *Ann Arbor, MI*

- Trained a **conditional flow-matching** generative model over a multi-regime flow manifold, steering its sampling trajectories with **Terminal Reward Guidance** to synthesize physically realizable, reward-maximizing targets.
- Trained a lightweight **deep-RL policy** to track those targets, reaching **48.95% drag reduction** at $Re_\tau = 180$ (**+16.2% over the DRL benchmark**) at **37× lower actuation energy**.
- Built the end-to-end **PyTorch/GPU** stack over 22,000+ DNS snapshot pairs; the deployed controller runs with **no generative queries at inference**.

CFD Foundation Models for Geometry-Generalizable Flow Prediction Jul 2026 – Present
Graduate Researcher, University of Michigan *Ann Arbor, MI*

- Developing a **pretrained CFD surrogate** that generalizes across unseen geometries without per-case retraining, pairing a **joint-embedding predictive architecture (JEPA)** with a mesh-density-preserving geometry encoding.
- Preliminary results show **zero-shot prediction on held-out geometries** without fine-tuning, targeting inference-time flow estimation at a fraction of solver cost.

Deep Learning for Fluid Mechanics - Research & Teaching Assistant Jan 2025 – Present
University of Michigan · Flowthermolab (industry collaboration) · Siemens (Stockholm) *Ann Arbor, MI*

- Designed and taught a hands-on deep-learning curriculum (MLPs, RNNs, autoencoders, CNNs) for 30+ graduate students and practicing engineers.
- Authored the course's core **PyTorch/Jupyter** notebook suite on **flow-field reconstruction, reduced-order modeling, and data-driven turbulence closure**; delivered the applied version to Siemens engineers in Stockholm.

Upstream Pressure-History Effects on Adverse-Pressure-Gradient TBLs Jun 2024 – Jun 2025
Advisor: Prof. Ricardo Vinuesa, KTH Royal Institute of Technology *Stockholm, Sweden*

- Analyzed **DNS/LES** campaigns on turbulent boundary layers under varying upstream pressure histories, quantifying non-local memory effects via spectral densities, Reynolds-stress budgets, and β-evolution. Published in *Int. J. Heat and Fluid Flow* (2025).

GPU-Accelerated Meshfree CFD Solver (CUDA) Nov 2022 – Dec 2023
Advisor: Dr. Anil Nemili, BITS Pilani *Hyderabad, India*

- Built a 3D meshfree Euler solver in Fortran/CUDA and drove a **450× GPU speedup** via kernel restructuring, memory-coalescing, and NVIDIA Nsight-guided profiling.

Uncertainty-Aware Airfoil Pressure Reconstruction from Sparse Taps - ***Conditional Flow Matching*** Jan 2026 – Apr 2026

- Built a **conditional flow-matching** model reconstructing full airfoil surface pressure from as few as **5–20 pressure taps** with calibrated uncertainty, pairing a **DeepSets encoder with learned positional attention** with a 1D conv UNet vector field (FiLM-conditioned at every scale). Code
- Reached **183 Pa MAE** (14% of pressure std) at 10 taps on **AirfRANS**, recovering C_L/C_D to $\Delta C_L = 0.094$; posterior is well-calibrated at **97% coverage** within $\pm 2\sigma$.

Professional Experience

Applied Aero Labs Pvt Ltd - ***Co-founder & Technical Lead*** Apr 2023 – May 2025

- Co-founded an aerodynamic R&D venture building India's first portable, multi-velocity fan-array wind tunnel; secured **$12,000 in government funding** and owned mechanical design, embedded control/electronics integration, and client validation.

Leadership & Teaching

President, SEDS BPHC *(2023)*: led the technical roadmap of a 120-member student space org. **Structures Lead, Hyperloop India**: pod-frame and vacuum-seal design. **TA, Fluid Mechanics**: 200 undergraduates; tutorials and CFD labs.